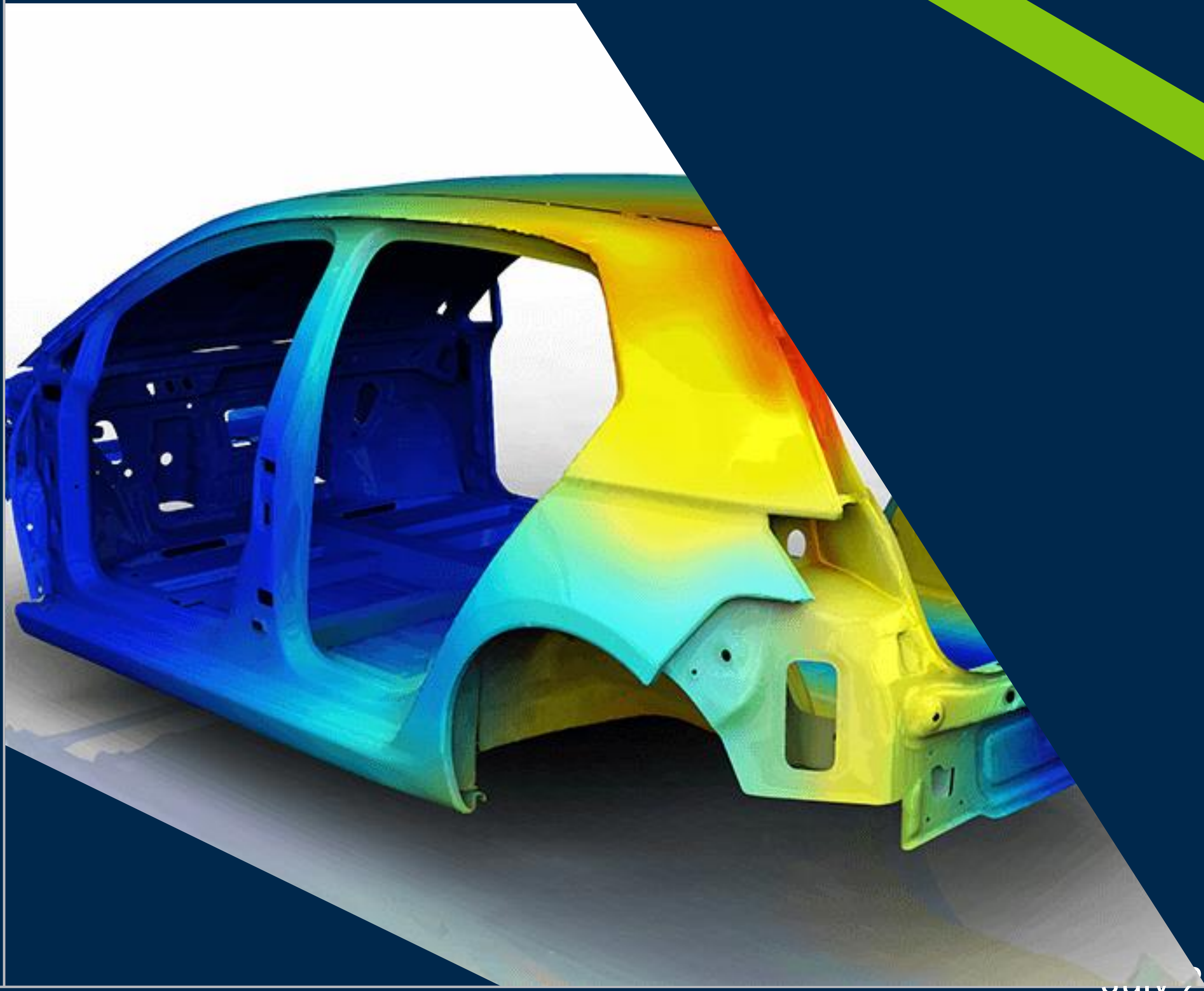
HEXAGON

MSC Nastran



Release 2025.2:

What's new

July 2023

Introduction to MSC Nastran

MSC Nastran | Multidisciplinary structural analysis solver that performs static, dynamic, and thermal analysis across the linear and nonlinear domains

- Simulate and test designs virtually to reduce costly physical prototypes
- Reduce reliance on physical testing in extreme or hazardous environments
- Remedy structural issues that may occur during a product’s service, reducing downtime and costs.
- Optimise the performance of existing designs or develop unique product differentiators to gain a competitive advantage
- Certify designs in the aerospace, automotive, and shipbuilding industries
- Take advantage of Hexagon’s interoperability to co-simulate across multiple disciplines.

MSC Nastran is a high-performance FEA solver for linear and nonlinear analysis with an open framework that includes multiphysics and model-based systems.

HEXAGON

Release summary

MSC Nastran 2025.2

Release highlights

Major updates and enhancements in MSC Nastran 2025.2

New capabilities:

- A new linear supersonic panel method called Mach1+ has been added.
- The linear gap in SOL 101 simulates compressive-only contact between two parts.
- Frequency-dependent DMIG rotor matrices and 6x6 BUSH elements allow speed-scalable data entry for rotordynamic analysis.
- Co-simulate MSC Nastran and Adams using MSC CoSim.

Enhancements:

- Monitor points provide a summation of a set of grid point forces for random vibration analysis.
- Optimise the dynamic frequency response directly in the assembly with FRDISP constraints.

Enhancements:

- The Multifrontal Massively Parallel Solver (MUMPS) is integrated to support frequency and transient response analyses.
- The nonlinear solver (NLPERF) now supports nonlinear transient analysis and new convergence criteria.

New capabilities and enhancements in MSC Nastran 2025.2

Linear gap (LGAP)

Description:

Linear gap modelling (LGAP) uses CGAP elements to simulate compressive-only contact between two parts, assuming no friction, linear elastic materials, and negligible large displacement. It enables point-to-point contact modelling in linear analysis scenarios, such as press-fit or fitting contacts, hole-bearing, and heel-to-toe contact. The linear gap allows for activation and deactivation (point-to-point) of local contacts, capturing only the nonlinear elements of the problem without requiring a full nonlinear solver.

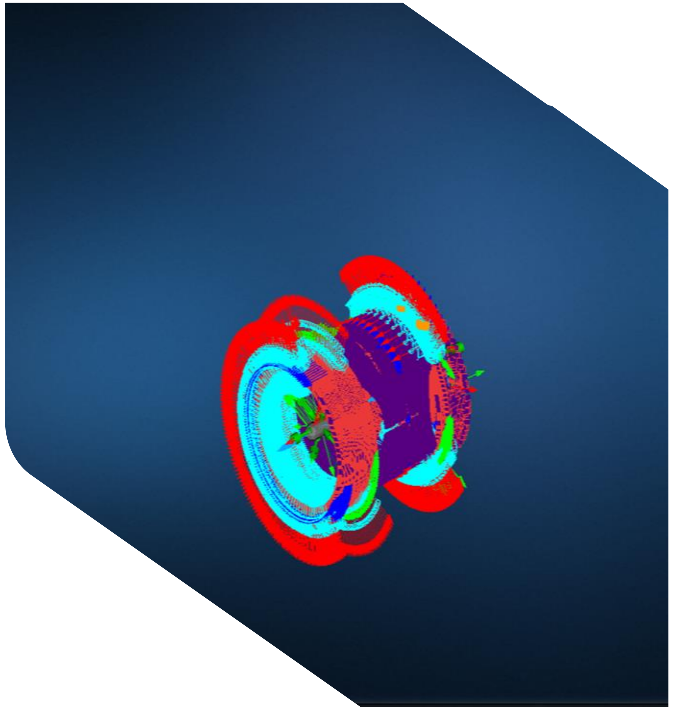
Benefits:

- Contact analysis is more efficient when there are limited nonlinear elements, and runs are faster.
- Users do not need to purchase a standalone nonlinear solution.
- Engineering joints and assembly fit-up simulations are easier to run because contact can be activated or deactivated locally.

Who needs it:

- Users in the aerospace industry

The example shown is a mechanical device under thermal loading. The red arrows around the perimeter are gap element orientation.



New capabilities and enhancements in MSC Nastran 2025.2

Mach1+ supersonic aerodynamics panel method

Description:

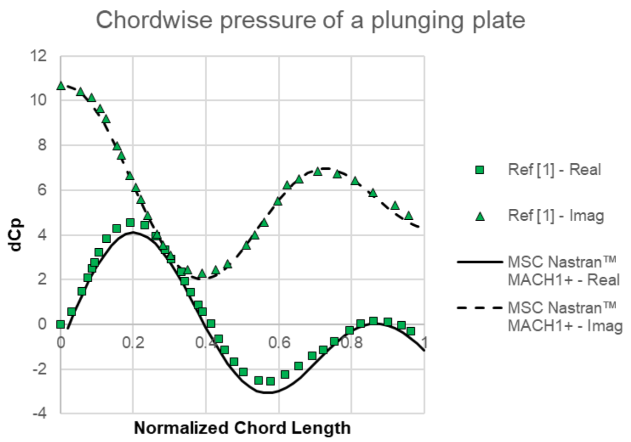
MSC Nastran 2025.2 introduces Mach1+, a new modern linear supersonic panel method for aerodynamic simulations. Mach1+ is directly integrated into the key aeroelastic and flutter solution sequences (SOL 144, 145, 146, and 200) of MSC Nastran and uses identical bulk data entries and user experience as the previous MSC Nastran versions.

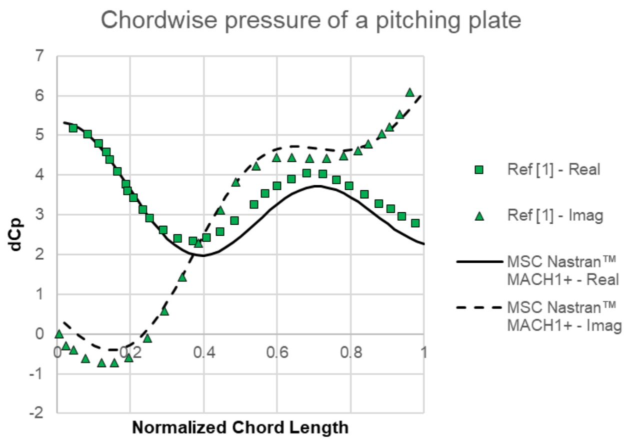
Benefits:

- Problems can be solved entirely in MSC Nastran, without relying on other solutions.
- Seamless migration means no changes required to the existing supersonic aerodynamic input files.
- The solution is thoroughly validated for accuracy through academic and industrial user cases.

Who needs it:

- All users in the aerospace industry





New capabilities and enhancements in MSC Nastran 2025.2

MONPNT3 support for random vibration analysis

Description:

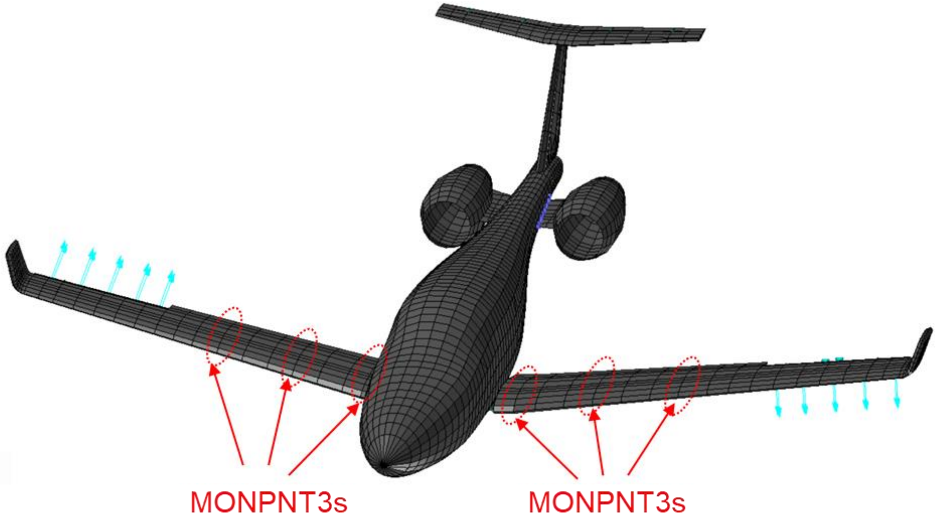
Monitor points (MONPNT3) in MSC Nastran provide a summation of a set of grid point forces at specific locations in a specific coordinate system. MONPNT3 is now supported in MSC Nastran random vibration analysis. This allows users to obtain key random vibration quantities, such as Power Spectral Density (PSD), Root Mean Square (RMS), Number of Zero Crossings (N0), and Cumulative RMS (CRMS) at defined monitor points.

Benefits:

- Post-processing capabilities are enhanced with more options.

Who needs it:

- All users in the aerospace industry



HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

Frequency-dependent DMIG rotor matrices

Description:

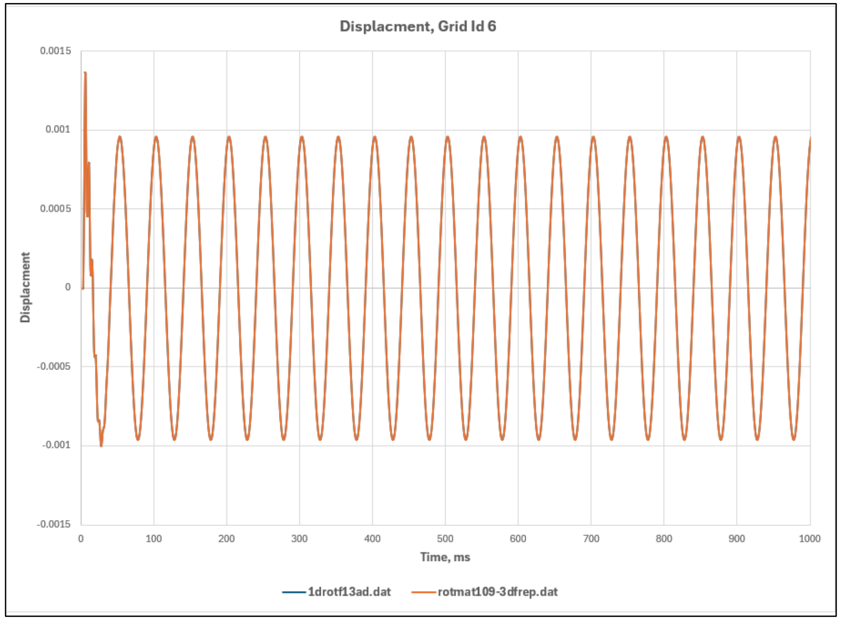
The new ROTMAT bulk data entry enables input of rotor matrices in DMIG format in MSC Nastran, allowing speed-scalable data entry for rotordynamic analysis. Specify mass, stiffness, damping, gyroscopic, and circulation terms calculated at the unit speed, which will be scaled by the speed/frequency of interest. Bulk data entry supports eigenvalue extraction and unbalanced response in linear and nonlinear frequency and time domains.

Benefits:

- Define rotor properties using matrices instead of traditional finite elements to enhance the modelling flexibility of complex submodels.
- Collaborate by sharing rotor models in DMIG format while protecting proprietary or sensitive finite element information.
- Automatically scale rotor matrices from unit speed to operating speed to simplify analysis and reduce errors.

Who needs it:

- All users doing rotordynamic analysis



HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

Frequency-dependent 6x6 BUSH elements

Description:

MSC Nastran 2025.2 introduces expanded capabilities for BUSH elements, allowing users to define full 6x6 stiffness and damping matrices for joints and connections. Using new identifiers, engineers can specify static (nominal) or frequency-dependent 6x6 matrices directly in PBUSH or PBUSHT entries. This enables precise simulation of connections (flexible joints, bushings, rotor supports) with non-symmetric or cross-coupled physical properties across all six degrees of freedom.

Benefits:

- Users can simulate real-world connection behavior (such as flexible joints and bushings) much more closely, leading to better predictions of system response.
- Model full 6-DOF support via the BUSH element, resulting in greater simulation accuracy for non-symmetric or cross-coupled physical properties across all six degrees of freedom.

Who needs it:

- Aerospace engineers simulating rotor dynamics, flexible supports, and advanced joints, and any other industry where the full 6-dof stiffness and damping are needed
- While most useful for aerospace rotordynamics, this capability can be applied to any industry that needs full 6-DOF stiffness and damping capabilities.

New capabilities and enhancements in MSC Nastran 2025.2

Double-Sided Blade (DSB) stiffener support

Description:

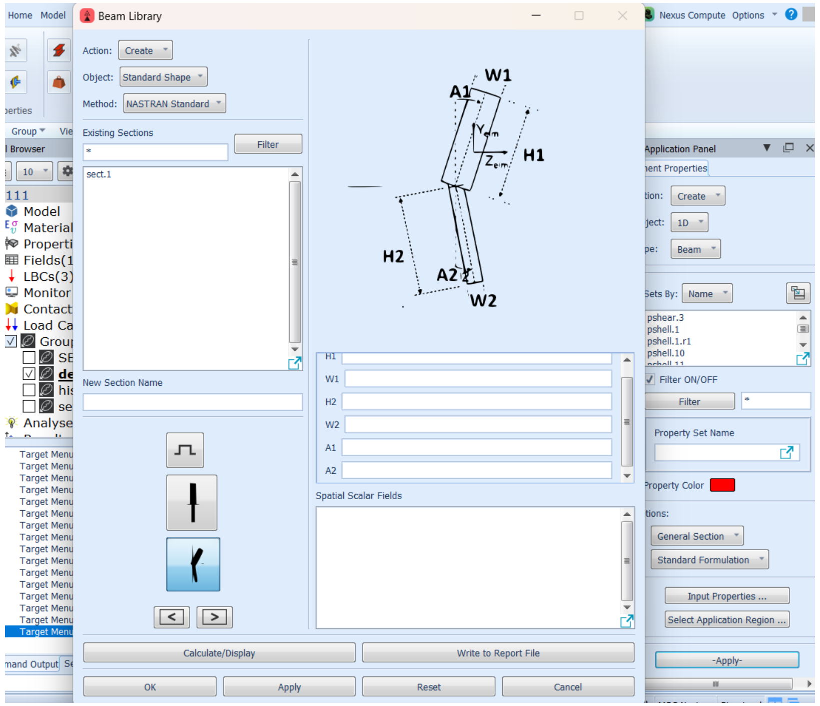
MSC Nastran 2025.2 introduces a new, predefined cross-section for Double-Sided Blade (DSB) stiffeners in the PBARL and PBEAML bulk data libraries. The DSB stiffener allows a blade on each side of the main web to have independent height, thickness, and angles. Both blades can differ in dimensions and orientation (angle from vertical), enabling accurate modelling of advanced stiffening problems in aerospace structures. Sizing optimisation is also supported. This is integrated into both PBARL and PBEAML.

Benefits:

- Get direct support for double-sided blade stiffeners not available in other commercial tools.
- Customise both blades with different heights, thicknesses, or angles for more precise modelling and optimisation.

Who needs it:

- All users in the aerospace industry



New capabilities and enhancements in MSC Nastran 2025.2

Frequency-dependent scale factor capability for the PBUSHT

Description:

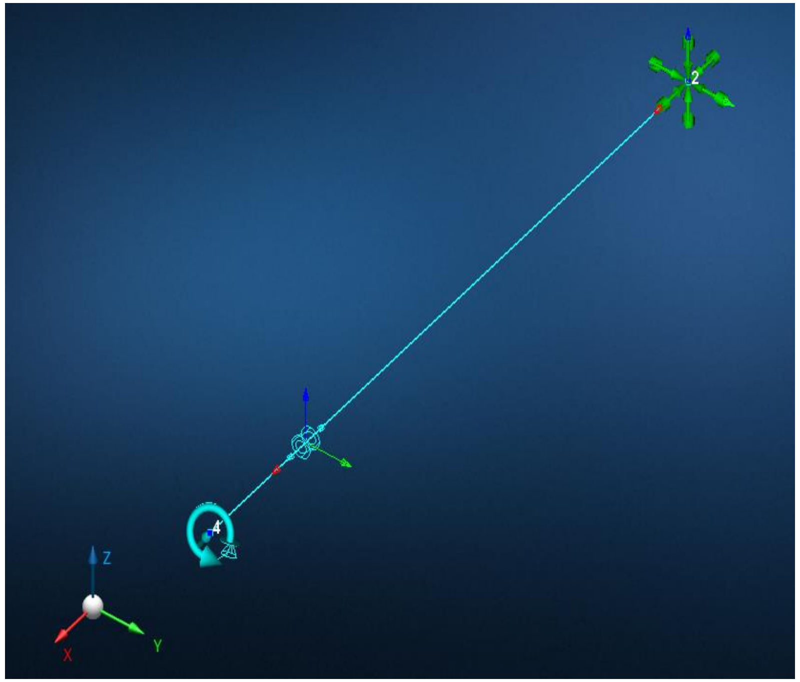
MSC Nastran 2025.2 enables frequency-dependent scaling of bushing element properties using three new keywords on the PBUSHT entry: KSCAL (stiffness), BSCAL (damping), and GESCAL (structural damping). These scale factors allow users to link static nominal bushing properties (defined in PBUSH) to frequency-dependent values from tables, resulting in frequency-adaptive simulation behavior for bushings.

Benefits:

- Only update the nominal value when bushing properties change.
- Maintain frequency dependency relationships automatically.
- Reduce the potential for errors during multiple iterations.

Who needs it:

- All users



Simple cantilevered beam with CBUSH positioned 80% of the beam length toward the tip.

New capabilities and enhancements in MSC Nastran 2025.2

MUMPS in frequency and transient response

Description:

The Multifrontal Massively Parallel Solver (MUMPS), is integrated into MSC Nastran 2025.2 to support frequency and transient response analyses. It enhances solution scalability and performance in SMP environments, supporting direct and modal solution sequences (108, 109, 111, 112), design optimisation (SOL 200), and linear transient analyses (SOL 400).

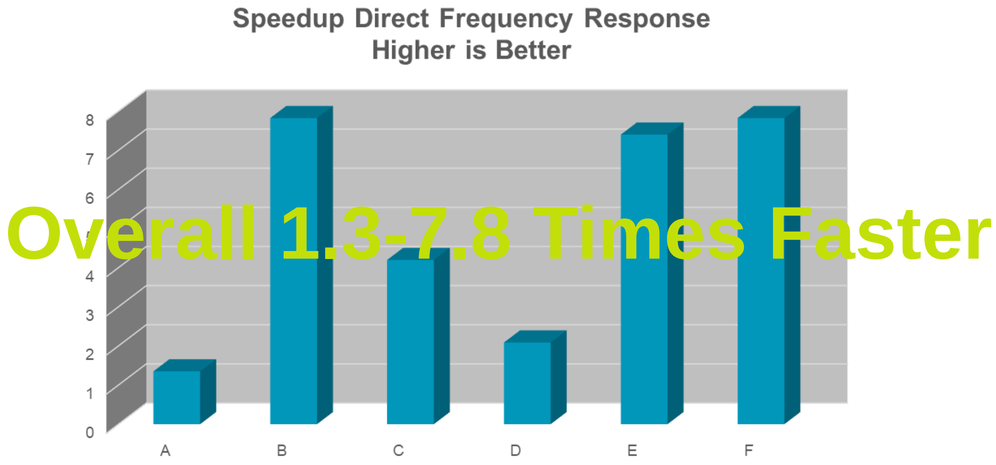
Benefits:

- This is an efficient, scalable solution for large frequency and transient response models.

Who needs it:

- All users

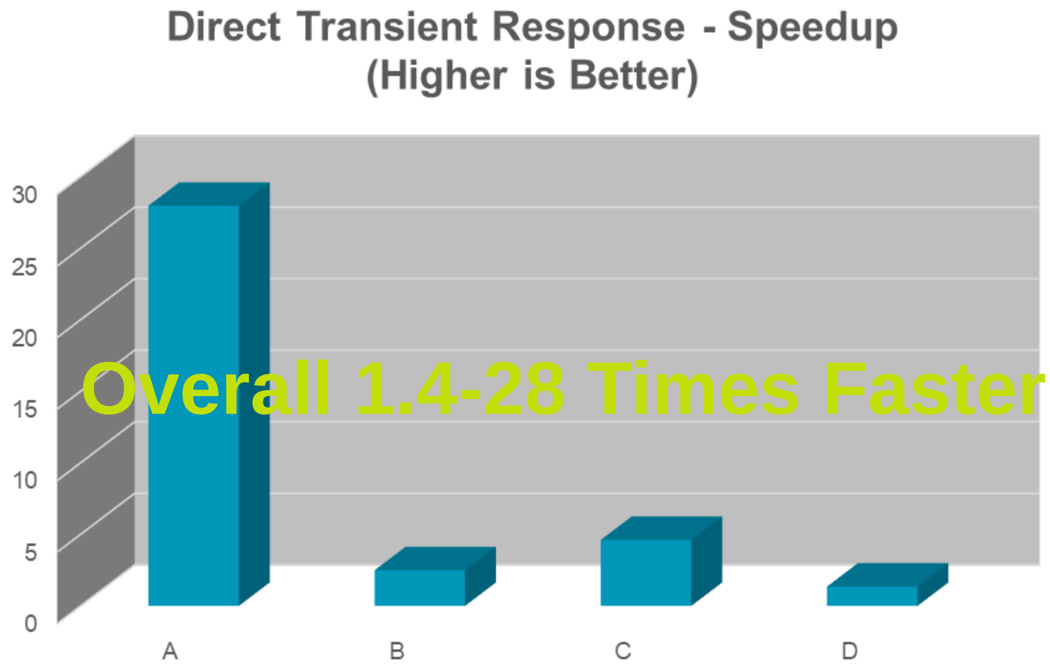
Frequency Response Test Models			
MODEL	DOF	Frequencies	**Form**
A	19.7m	6	sym
B	19.7m	6	unsym
C	1.4m	2	Sym
D	6.9m	1	Sym
E	7.4m	1	Sym
F	19.7m	6	Sym



HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

MUMPS in frequency and transient response (continued)



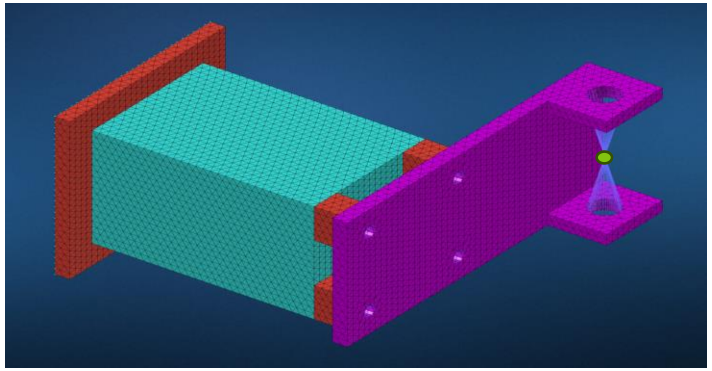
Transient Response Test Models			
MODEL	DOF	Time Steps	Form
A	17.7m	71	Unsym
B	0.5m	100k	Unsym
C	6.4m	8135	Unsym
D	33.2m	3000	Sym
E			

New capabilities and enhancements in MSC Nastran 2025.2

SOL 200 NEO maximum displacement constraints in assembly contexts

Description:

MSC Nastran 2025.2 allows users to place a maximum displacement constraint at any arbitrary point in an assembly. RBE connections aren't needed. Any point in the assembly—including locations outside the design space—can now be directly constrained.

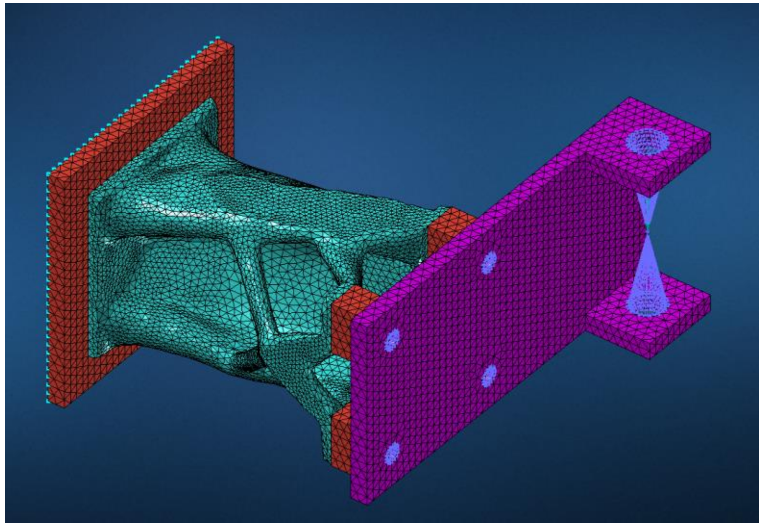


Benefits:

- Optimize a single component design but influence full assembly behaviour.
- Enjoy improved robustness, effectiveness, performance, and usability when using SOL 200 NEO.

Who needs it:

- All users doing topology optimization analysis



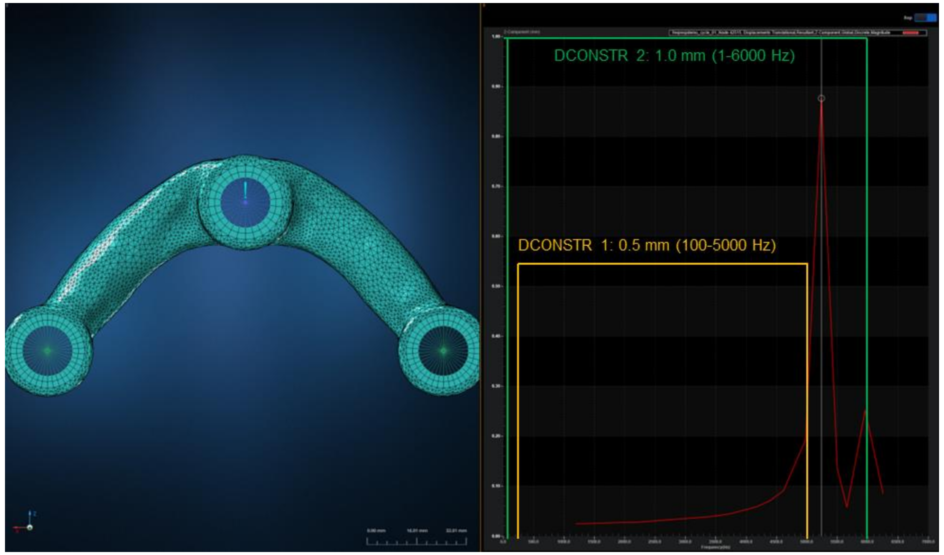
HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

SOL 200 NEO - dynamic frequency response with FRDISP constraints

Description:

MSC Nastran 2025.2 now fully supports dynamic frequency response optimisation directly in the assembly context via FRDISP constraints. This first release only supports FRDISP as a constraint, and the constraint value must be chosen carefully, and must fit the model to receive suitable designs.



Benefits:

- Extend SOL200 NEO to frequency response problems.

Who needs it:

- All users doing topology optimization analysis

New capabilities and enhancements in MSC Nastran 2025.2

AVL EXCITE™ interface: separate structural damping matrix

Description:

AVL EXCITE™ is widely used to simulate rigid and flexible multibody dynamics (MBD) in automotive powertrains. MSC Nastran offers end-to-end support of AVL EXCITE by exporting EXB files directly and recovering data in MSC Nastran from an AVL EXCITE simulation. Previously, only an equivalent viscous damping matrix was exported, which combined several damping sources (viscous, structural, Rayleigh, modal). The new feature allows the structural damping matrix to be output independently, as a separate matrix in the EXB file.

Benefits:

- This feature provides greater flexibility and fidelity in modelling damping behaviour when simulating the AVL Flexbody.

Who needs it:

- Automotive engineers performing multibody dynamics (MBD) powertrain simulations with AVL EXCITE™.

```
$ File Manage Management Section
ASSIGN OUTPUT2='crankextse.op2',UNIT=80,DELETE
$ Exec. Control Section
SOL 103
$ If required used ACMS
DOMAINSOLVER ACMS (VERSION=NEW)
CEND
$ Case Control Section
TITLE = Sample Nastran Input File: Step 1
$ AVLEXB Case Control Above Subcase Level
$ Export EXB Flexbody with Full Mass Invariants
$ Partial Recovery Matrix involving user defined set U2
$ Specify DAMPSEP = YES to output Separate Structural Damping
AVLEXB EXBBODY=YES DAMPSEP = YES
METHOD = 103
...
$ To minimize the size of External Superelement
$ .op2 the user should only request outputs for
$ sets of physical quantities that of interest.
$ For, e.g., displacement and velocities on
$ surface nodes
EXTSEOUT(ASMBULK,EXTBULK,EXTID=20) DMIGOP2=80
...
$ Nodal set 100 and Element set  102 should be defined
DISP(PLOT)=100
$
STRESS(PLOT)=102
```

New capabilities and enhancements in MSC Nastran 2025.2

Air flow resistivity default change in PEM materials

Description:

The convention for Air Flow Resistivity (AFR) in porous materials has changed in MSC Nastran 2025.2. Previously, AFR was measured using only the flow through the fluid phase. The new convention measures AFR using the total flow going through the entire porous material, not just the fluid phase.

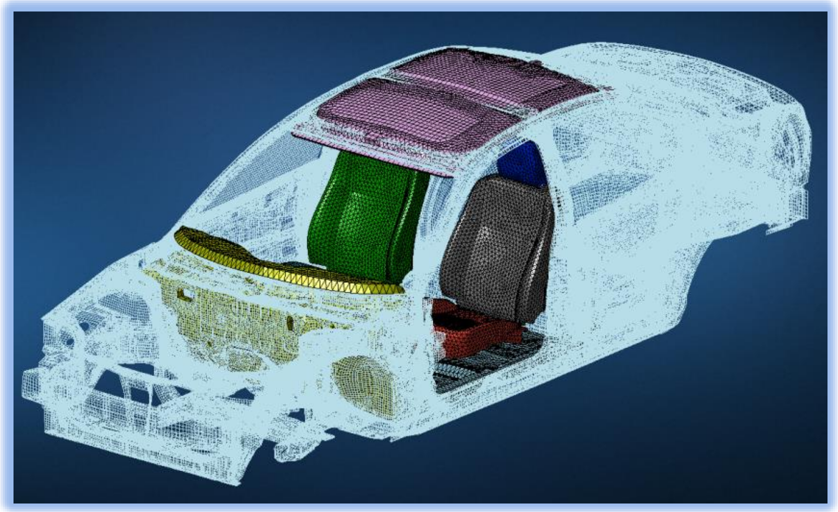
- This change of convention means that results obtained with MSC Nastran 2025.2 will not match results obtained with prior versions.
- To keep the old convention, the AFRBC flag must be set to 1 in all MATPE1 bulk data entries.
- To use the new convention, no change is required

Benefits:

- Get a more representative and comprehensive approach to modelling porous materials in PEM analysis.
- Use the most common conventions for PEM modelling.

Who needs it:

- All users doing PEM analysis in MSC Nastran



MATPE1 bulk data entry is extended

1	2	3	4	5	6	7	8	9	10
MATPE1	MID	MAT1	MAT10	BIOT	POROPT	SRHO	AFRBC		
	VISC	GAMMA	PRANDTL	POR	TOR	AFR	VLE	TLE	

HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

Nonlinear transient analysis with NLPERF

Description:

The recently introduced SOL400 NLPERF solver has delivered significant performance gains compared to the classic SOL400 solver (CNLS) for nonlinear static (NLSTAT) analysis.

Building on this success, the 2025.2 release extends similar improvements to nonlinear transient dynamic (NLTRAN) analysis, enabling efficient simulation of structures under dynamic loads and accurate prediction of responses to nonlinear transient effects.

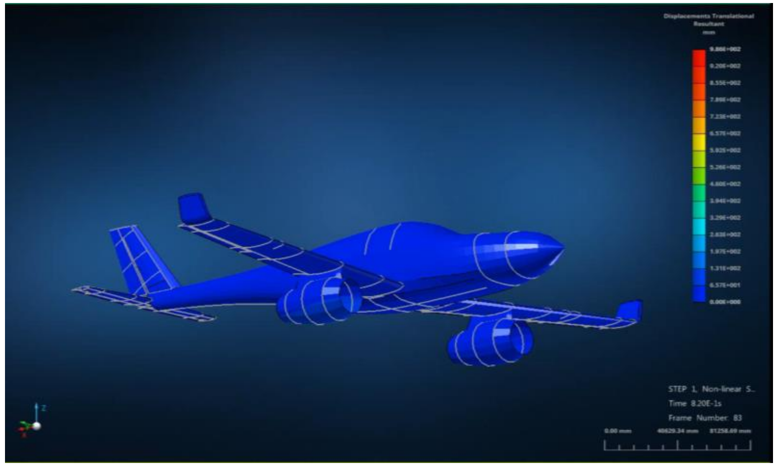
To achieve these enhancements, NLPERF now supports:
- Comprehensive mass and inertia elements (CONMi, CMASSi, NSM, …)
- All damping types and elements (viscous, structural, CBUSH, etc.)
- All dynamic loading types (TLOADi, NONLINi, NLRGAP, NLRSFD, …)
- Initial condition options (ICOPT = 0, 1)
- Multiple time integration algorithms (SSH, HHT, Newmark-Beta, …)
- Expanded output options.
- Analysis chaining: nonlinear static then nonlinear transient dynamic.

Benefits:

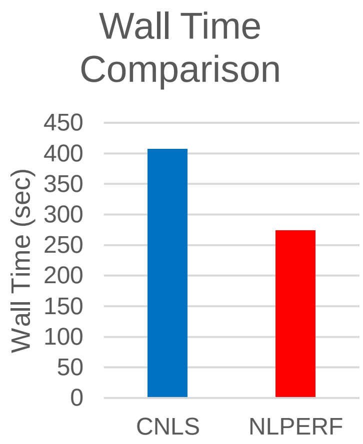
- Reduce runtimes for transient analysis compared to the legacy CNLS solver.

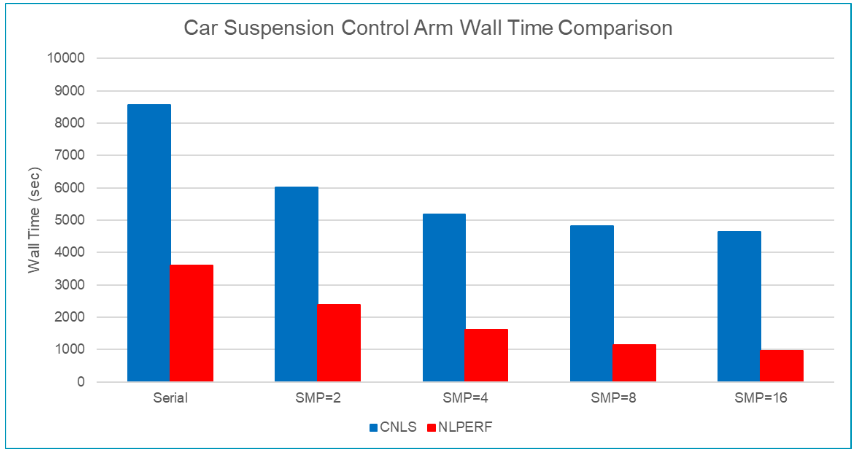
Who needs it:

- All users



Displacements during Engine Breakaway





Wall-time comparison for a car suspension control arm

HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

New convergence criteria and method available in NLPERF

Description:

The weighted norm-based convergence method, originally available in the legacy CNLS solver, uses weighted norms instead of simple maximum component checks and can now be used with the NLPERF solver. This approach evaluates convergence based on an average error across the entire model, weighted appropriately, rather than focusing on the largest error at a single degree of freedom.

This method complements the existing options for convergence checks in NLPERF based on force/moment residuals (P) and/or displacements/rotations (U). Additionally, a new option for checking strain energy/work (W) has been introduced.
Supported convergence criteria combinations: U, P, UP, UW, PW, and UPW.
By default, or when W is specified alone, NLPERF will switch to UPW if no contact is defined in the model, or to PV in other cases.

Benefits:

- Achieve greater control over convergence criteria.
- Weighted norm-based checks using incremental displacement and work energy provide a more balanced and less restrictive convergence assessment.
- Load (P) convergence is performed using a weighted norm of external forces, reducing convergence issues in models without SPCs.

Who needs it:

- Advanced users who want full control over convergence criteria

Parameter	CNLS	NLPERF (2025.1)	NLPERF (2025.2)
CONV='blank'	UPW (non-contact), PV(contact)	PV	UPW (non-contact), PV(contac
CONV=UW,PW,UPW	UW,PW,UPW	UV,PV,UPV	UW,PW,UPW
CONV=U, P, UP	U, P, UP with weighted norm	UV, PV, UPV	U, P, UP with weighted norm
EPSU (>0.0)	Incremental displacement (NLSTEP) Total displacement (NLPARM, TSTEPNL)	Incremental displacement	Incremental displacement
EPSU (<0.0)	Incremental displacement	Incremental displacement	Incremental displacement
EPSP	Total external/reaction force	Total external/reaction force	Total external/reaction force
EPSW (>0.0)	Total energy	Incremental energy	Incremental energy
EPSW (<0.0)	Incremental energy	Incremental energy	Incremental energy

Convergence criteria used according to the parameter in the bulk file

```
Convergence Options (CONV) .............       UP
Intermediate Output Flag (INTOUT) .......            1
             - Displacement   (EPSU) ......    -1.00E-02
Tolerance    - Residual Force (EPSP) ......     1.00E-02
             - Work           (EPSW) ......    -1.00E-02
```

F06 file abstract with NLPARM, PFNT, UP, 1

HEXAGON

New capabilities and enhancements in MSC Nastran 2025.2

New energy outputs available in NLPERF

Description:

New global energy outputs have been introduced to enhance energy-based validation and model diagnostics:

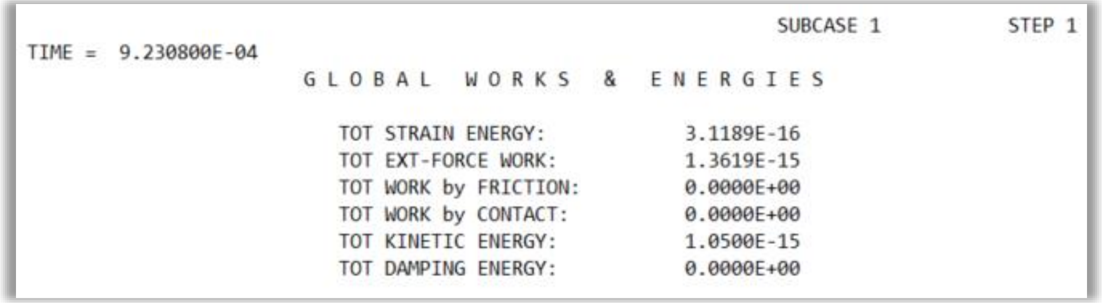
- A comprehensive summary of global energy balance for each converged load increment and at the end of the analysis, for both nonlinear static and transient analyses.
  - Total strain energy, total external work, work by contact forces, work done by frictional forces, total kinetic energy, total damping energy
- Results are provided in all standard output formats: F06, PCH, H5, OP2, and ITR. Future releases of MSC Apex and Patran will support these outputs.

Benefits:

- Get energy-based validation of nonlinear solutions for improving user confidence in simulation results.
- Simulation model debugging is easier due to clear energy flow information.

Who needs it:

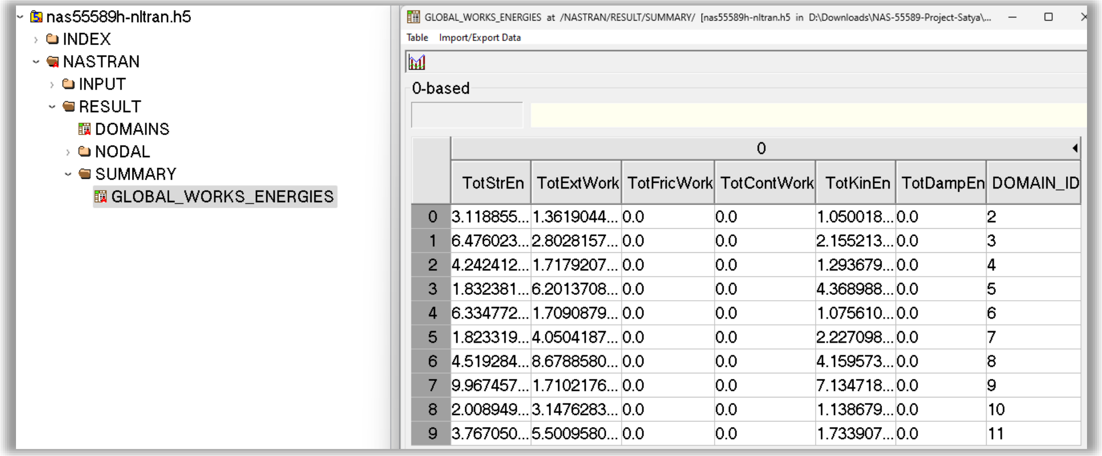
- Users requiring energy balance checks for model validation and troubleshooting.
- Teams working on contact-heavy or frictional models where energy tracking is critical



```
                                                          SUBCASE 1          STEP 1
TIME =   9.230800E-04
                        G L O B A L   W O R K S   &   E N E R G I E S

                           TOT STRAIN ENERGY:             3.1189E-16
                           TOT EXT-FORCE WORK:            1.3619E-15
                           TOT WORK by FRICTION:          0.0000E+00
                           TOT WORK by CONTACT:           0.0000E+00
                           TOT KINETIC ENERGY:            1.0500E-15
                           TOT DAMPING ENERGY:            0.0000E+00
```

Global energy balance available in f06 files



	TotStrEn	TotExtWork	TotFricWork	TotContWork	TotKinEn	TotDampEn	DOMAIN_ID
0	3.118855...	1.3619044...	0.0	0.0	1.050018...	0.0	2
1	6.476023...	2.8028157...	0.0	0.0	2.155213...	0.0	3
2	4.242412...	1.7179207...	0.0	0.0	1.293679...	0.0	4
3	1.832381...	6.2013708...	0.0	0.0	4.368988...	0.0	5
4	6.334772...	1.7090879...	0.0	0.0	1.075610...	0.0	6
5	1.823319...	4.0504187...	0.0	0.0	2.227098...	0.0	7
6	4.519284...	8.6788580...	0.0	0.0	4.159573...	0.0	8
7	9.967457...	1.7102176...	0.0	0.0	7.134718...	0.0	9
8	2.008949...	3.1476283...	0.0	0.0	1.138679...	0.0	10
9	3.767050...	5.5009580...	0.0	0.0	1.733907...	0.0	11

Global energy balance available in h5 files

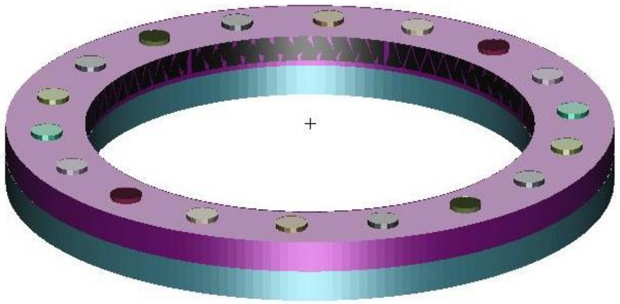
New capabilities and enhancements in MSC Nastran 2025.2

New contact output options available in NLPERF

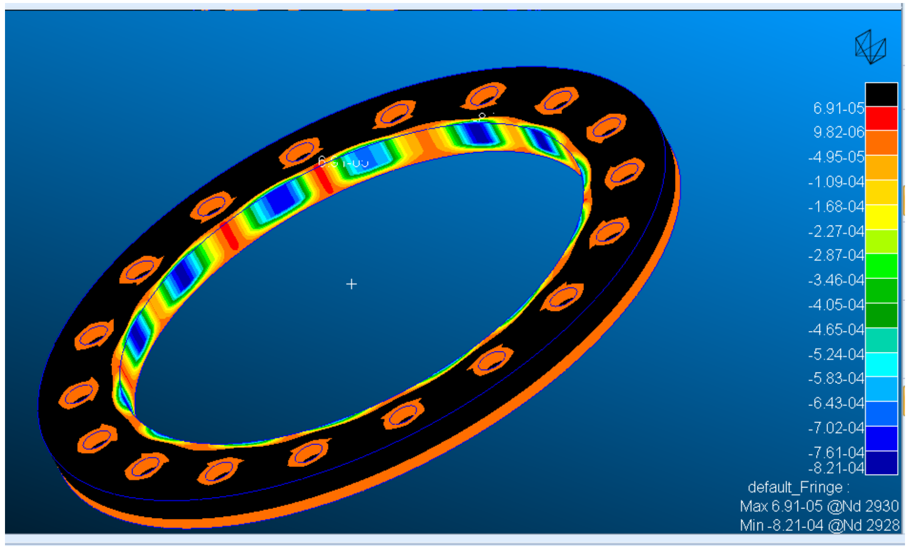
Description:

MSC Nastran 2025.2 introduces enhanced contact output capabilities to provide deeper insight into contact status and evolution during nonlinear analyses. These outputs now include gap distance information, complementing the existing BCONCHK contact check outputs, where distance checks are limited to nodes active within the contact search tolerance.

Additionally, users can now request contact outputs for the entire model or specific contact bodies, offering greater flexibility and control over the data to be reported.



Bolted assembly model



Contact gap distance outputs of a bolted assembly analysis

Benefits:

- Get improved model validation and debugging with transparent and comprehensive contact data.
- Enhance confidence in simulation accuracy, especially for complex contact interactions.
- Get greater flexibility through customizable contact output requests for targeted analysis of critical regions.

Who needs it:

- All analysts performing nonlinear simulations involving contact, such as assemblies, joints, seals, or frictional interfaces.

New capabilities and enhancements in MSC Nastran 2025.2

Element formulation information

Description:

Starting with MSC Nastran 2025.2, output information is now available in the f06 results file when using the modern NLPERF nonlinear solver, providing deeper insights into the FEA assumptions used during analysis. Those enhancements include:

- Comprehensive element descriptions for easier identification.
- Property IDs, useful for many cases such as shells and composite structures.
- Explicit indication of element formulations used in the analysis, with additional details available in the documentation.
- Module support is included.

Benefits:

- Get an improved understanding of the analysis assumptions and modelling choices.
- Improve confidence in simulation accuracy by making solver behaviour more transparent.
- Facilitate model verification and troubleshooting through clearer and more complete output data.

Who needs it:

- Simulation engineers who require detailed visibility into element formulations, types, and other information to validate the accuracy of nonlinear analyses.

GROUP.SUBGR	ELEM TYPE	No ELEMS	PROP TYPE	PID (MOD)	TYPE	FORM
1.1	CQUAD(4)	16	THKSHELL (DCT,L)	3 (1)	ADV	U
2.1	CBEAM(2)	10	PBEAM (BEAM,LC)	1 (2)	ADV	U
3.1	CBEAM(2)	10	PBEAM (BEAM,LC)	2 (3)	ADV	U
4.1	CHEXA(8)	4	PSOLID	5 (4)	CONV	U
4.2	CHEXA(8)	60	PSOLID	4 (4)	CONV	U
5.1	CELAS1	2	PELAS	111 (1)	CONV	U
5.2	CELAS1	1	PELAS	112 (1)	CONV	U
5.3	CELAS1	1	PELAS	111 (2)	CONV	U
5.4	CELAS2	1	N/A	0 (2)	CONV	U
5.5	CELAS2	1	N/A	0 (3)	CONV	U

Number of Element-Groups Used: 5

Number of Element-SubGroups Used: 10

FORMULATION:
U: updated Lagrange

Example of the information table now available in the f06 file

New capabilities and enhancements in MSC Nastran 2025.2

Co-simulation with MSC Nastran and Adams

Description:

MSC Nastran 2025.2 introduces a new co-simulation capability with the multi-body dynamics solution Adams via the CoSim product, using the modern NLPERF Solver in SOL 400. In this workflow:

- Adams provides enforced motion (displacements and rotations) of the interface nodes.
- MSC Nastran receives these displacements and returns the reduced stiffness matrix and reaction forces to Adams. Optionally, reduced mass and damping matrices can also be exchanged.
- The CoSim engine ensures synchronised data transfer at every time step.

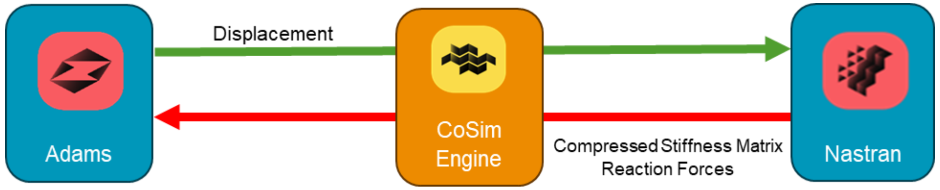
Currently, only nonlinear static analysis is supported on the MSC Nastran side. Support for nonlinear transient, dynamic analysis will be added in future releases.

Benefits:

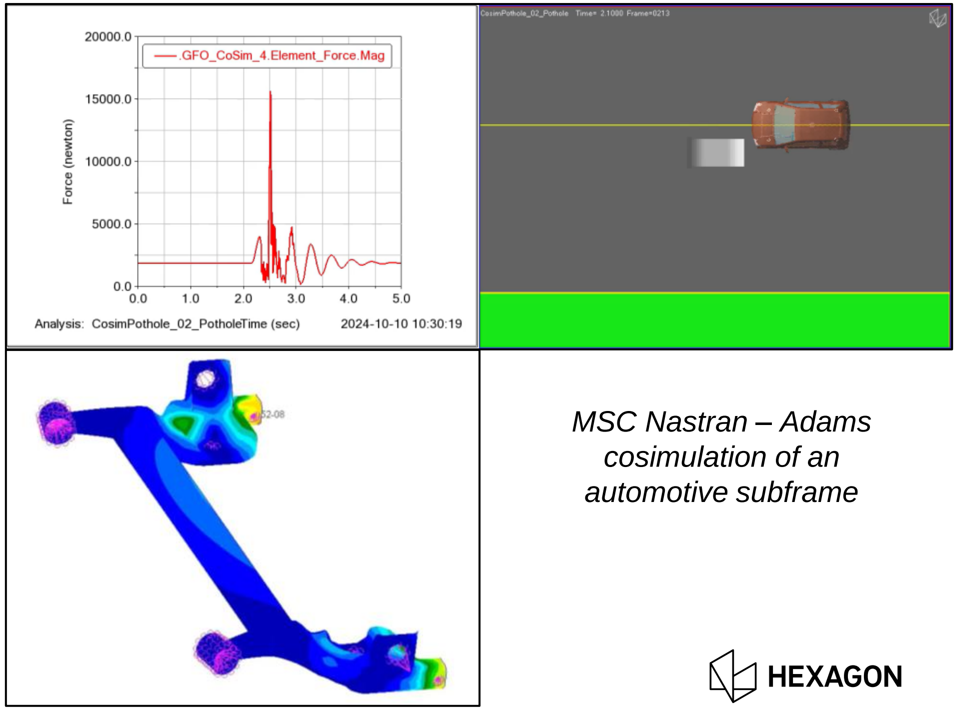
- Get a performant, integrated simulation of flexible structures and multi-body dynamics for more realistic system-level analysis.
- Improve accuracy and efficiency with synchronised solvers and reduce manual data exchange.
- Validate advanced designs for assemblies with complex motion and structural interactions.

Who needs it:

- Simulation engineers working on system-level analyses that combine structural flexibility with dynamic motion—such as automotive suspensions, aerospace mechanisms, robotics, and machinery



Graph introducing Adams – MSC Nastran co-simulation



MSC Nastran – Adams cosimulation of an automotive subframe

New capabilities and enhancements in MSC Nastran 2025.2

R-elements (e.g. RBE2, RBE3) enhancements

Description:

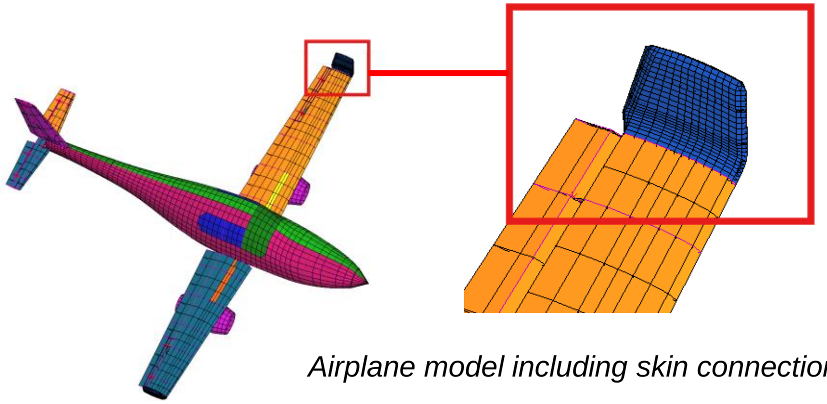
The R-Element Arbitrary User DOF Selection enhancement allows users to define custom degrees of freedom (DOFs) to be eliminated for R-elements beyond the previously available combinations. This capability provides greater flexibility in modelling complex connections, constraints, and coupling scenarios. Users can now specify which translational and rotational DOFs are active or constrained, enabling more precise representation of physical behaviour in multi-point connections or reduced models.

Benefits:

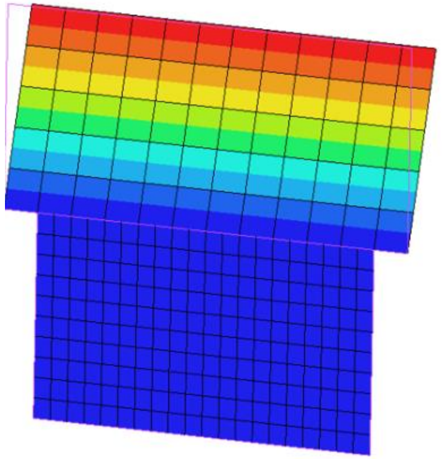
- Tailor R-elements to match real-world joint or interface conditions without unnecessary constraints for improved modelling flexibility.
- By selecting only relevant DOFs, users avoid over-constraining the model, leading to more realistic load paths and stress distributions.
- Use advanced techniques like component mode synthesis or co-simulation, where selective DOF activation is critical.

Who needs it:

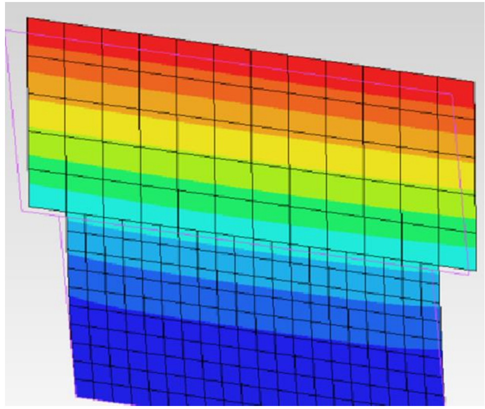
- Simulation engineers working on assemblies with complex joints.



Airplane model including skin connections



Continuous aircraft skin connection with non-aligning mid-side grid points



Hinged aircraft skin connection with non-aligning mid-side grid points showing more flexibility, so more deformation

Previous releases

MSC Nastran 2025.1, 2024.2 and 2024.1

Highlights from MSC Nastran 2025.1, 2024.2 and 2024.1

Adams .mnf for weakly coupled acoustics and improved performance:
MSC Nastran now exports fluid-structure interface matrices into the .mnf file, allowing users to recover acoustic pressures directly into Adams. The performance of mfn file creation has also been improved substantially with up to 4x speed-ups.

MUMPS support:
MUMPS solver support is available for Lanczos, residual vector, and buckling analysis. This upgrade delivers significantly faster, more reliable, and highly accurate results, helping users solve large problems with greater efficiency and confidence.

Single-step TPA:
Perform Single-Step Transfer Path Analysis (TPA) into MSC Nastran, increasing productivity and eliminating the need to maintain and combine separate dynamic analyses runs.

New NLPERF nonlinear solver:
MSC Nastran released a brand-new non-linear solver in SOL 400, achieving dramatic performance improvements and including advanced non-linear capabilities for complex non-linear workflows.

SOL 200 NEO for topology optimisation:
A brand new solver in SOL 200 Topology optimisation to output manufacturing-ready designs using generative design methodologies for component and full-assembly models.

Global-local analysis:
Detailed analysis of localised regions without needing to run the full model, saving both time and computational resources.

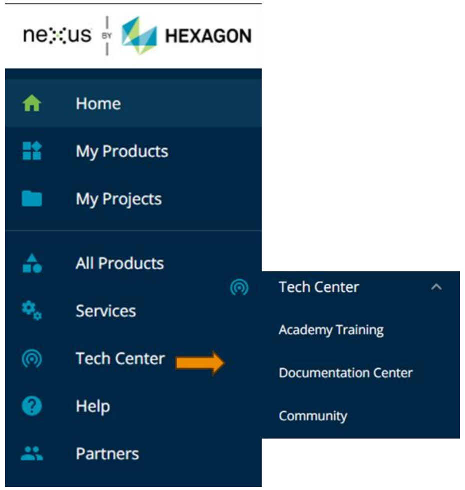
Additional resources

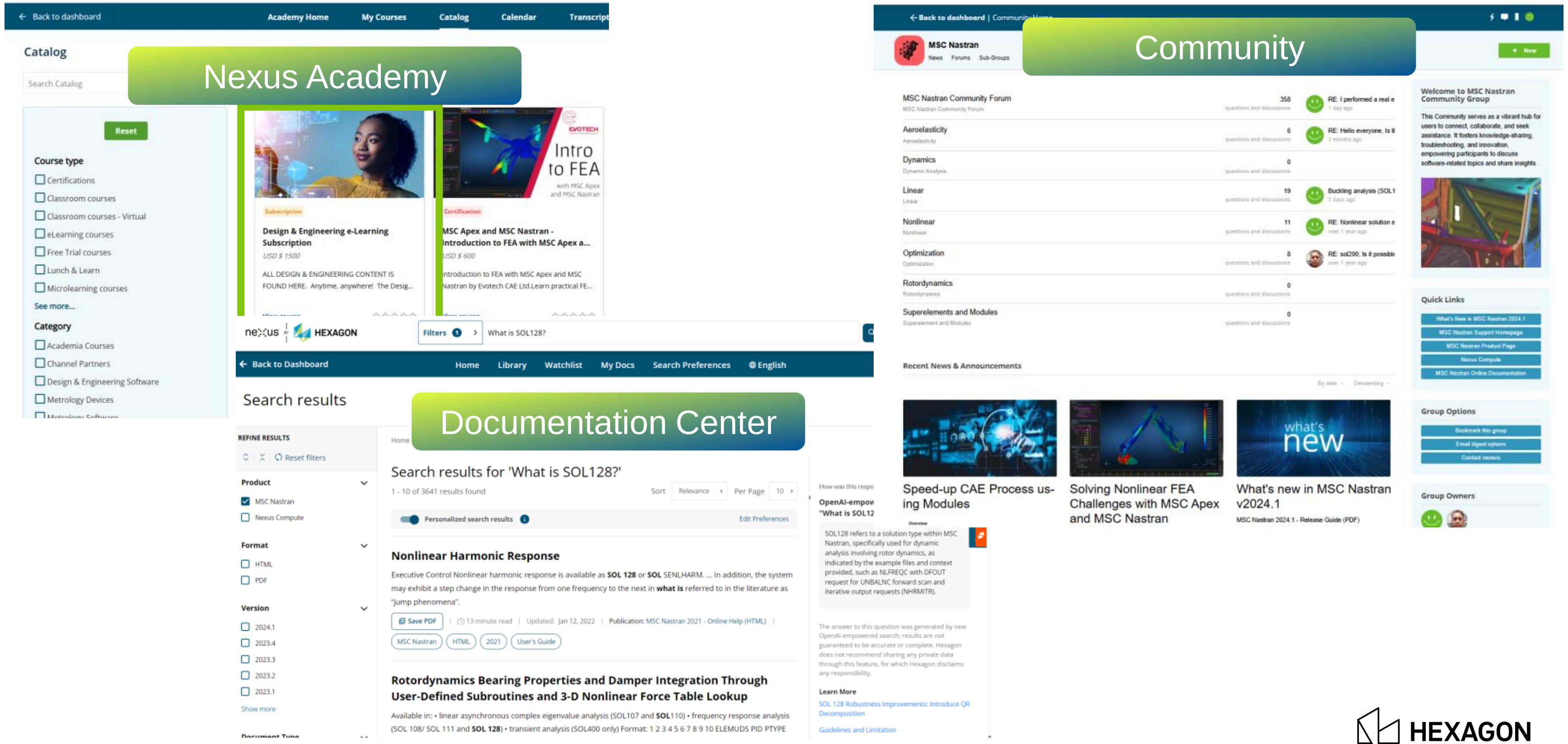
MSC Nastran 2025.2

Learn, search and ask for help

Nexus Home Page (https://nexus.hexagon.com/)

- **Free to register**
- MSC eLearning account migrated to **Nexus Academy** https://nexus.hexagon.com/home/hmi-learning-center-transition/
- SimCompanion Forums migrated to **Nexus Community** https://nexus.hexagon.com/home/simcompanion-forums-transition/





HEXAGON

Learn, search and ask for help

[Quick Access Links to MSC Nastran Resources]

SimCompanion, Hexagon.com

mscnastran.support@mscsoftware.com

msc_lic.support@mscsoftware.com